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BILLET-F4 DSP and Log-Mel Audio Frontend

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 03:14 UTC+8 Beijing Time)

Abstract

BILLET-F4 is a synthesizable four-channel audio frontend preserving programmable FIR and cascaded SOS filtering while adding rational polyphase resampling and a complete PCM-to-64-band Log-Mel pipeline. Its fixed feature profile uses 16 kHz audio, non-overlapping 512-sample Hann frames and unsigned 16-bit normalized acoustic tokens. Software-vs-Hardware numerical comparison established bit-exact agreement between independent Python mathematics and compiled RTL simulation. The principal PCM feature matrix covered 480 frames, comparing 123,360 spectral powers, 30,720 Mel energies and 30,720 top-level tokens. A programmed 48-to-16 kHz low-pass bank achieved 69.0628 dB stop-tone rejection and 0.0319 dB pass-tone amplitude error [E039]. L1 RTL timing lint passed without the prohibited hazards. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA closed the 10.0 ns clock with WNS_setup = +1.663 ns and hold slack = + 0.037 ns. L3 physical Arria 10 FPGA HIL passed three deterministic bridge-connectivity probes; these were register-access tests, not on-board BILLET-F4 workload comparisons [E044–E046].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top-level billet_f4_top retains the inherited filter hierarchy and selects one of four mutually exclusive data routes. Registered products, reductions, accumulators and rounding stages bound arithmetic depth. Serialized SRC, direct DFT and Mel engines favor straightforward numerical behavior and bounded storage over one-sample-per-cycle throughput. The spectrum engine reserves one 512-word frame buffer per channel; Mel reserves one 257-word power buffer. Embedded Hann, twiddle and compressed triangular-weight ROMs eliminate external coefficient-file dependencies. Independent histories and configuration banks isolate channels, while reservations prevent configuration changes from overtaking accepted work [E001, E002, E019].

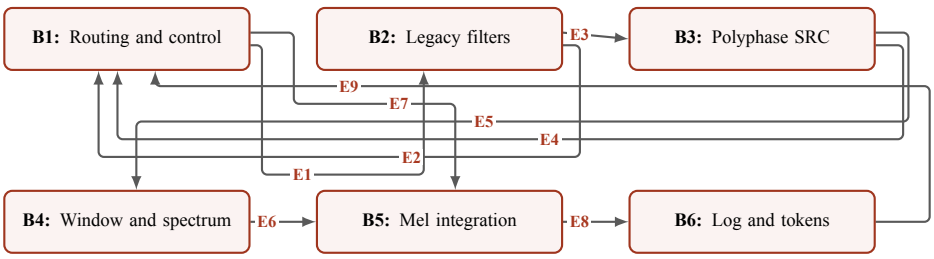


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Routing and control	billet_f4_top selects PCM or spectral ingress, arbitrates descriptors and tracks outstanding work. Backpressure and drain conditions preserve transaction ordering.

Block	Responsibility
B2: Legacy filters	u_legacy instantiates billet_f3_top with the inherited FIR and SOS hierarchy. Retimed arithmetic preserves transaction values, coefficient banking and history-clear rules.
B3: Polyphase SRC	u_src provides eight programmable phase slots per channel, double-banked coefficients and circular history. Registered compare/subtract scheduling supports interpolation and decimation.
B4: Window and spectrum	u_spectrum applies Hann weighting, reserves four independent frame buffers and dispatches complete frames round-robin through a serialized 512-point DFT.
B5: Mel integration	u_mel validates spectral framing, stores 257 powers and calculates 64 triangular weighted energies. Each bin's ROM entry represents at most two supported bands.
B6: Log and tokens	u_log performs piecewise-linear log2, fixed-point dB conversion, clipping and token normalization through eight globally stallable registers.

Directed relationship	Meaning
E1: B1 to B2	Accepted PCM and inherited configuration
E2: B2 to B1	Legacy PCM output in mode 0
E3: B2 to B3	Filtered PCM in modes 1 and 2
E4: B3 to B1	Resampled PCM output in mode 1
E5: B3 to B4	Resampled PCM in mode 2
E6: B4 to B5	Ordered spectral-power frames
E7: B1 to B5	Direct spectral ingress in mode 3
E8: B5 to B6	Energy with channel, band and last
E9: B6 to B1	Held token output in modes 2 and 3

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge synchronous operation with asynchronous active-low reset. Transfers use pre-edge valid && ready; reset forces all public outputs, including readiness, to zero.
s_valid, s_ready, s_data, s_channel	Input; s_ready output	1, 1, signed 16, 2 bits	PCM ingress for channels 0–3. Sources retain payload while stalled. Disabled in feature mode 3.
m_valid, m_ready, m_ data, m_channel	Output; m_ ready input	1, 1, signed 16, 2 bits	Legacy or resampled PCM output in feature modes 0 and 1. Valid and payload are zero in modes 2 and 3.
cfg_valid, cfg_ready, cfg_channel, cfg_addr, cfg_data	Input; cfg_ ready output	1, 1, 2, ADDR_W, signed 16 bits	Inactive FIR-bank writes. ADDR_W=max(1,clog2(TAP_COUNT)); coefficients use Q1.15.
commit_valid, commit_ ready, commit_channel	Input; commit_ready output	1, 1, 2 bits	Per-channel FIR bank swap, accepted only with complete staging coefficients and an eligible boundary.

Signal / group	Direction	Width	Contract
sos_cfg_valid, sos_cfg_ready, sos_cfg_channel, sos_cfg_section, sos_cfg_addr, sos_cfg_data	Input; sos_cfg_ready output	1, 1, 2, 2, 3, signed 18 bits	Inactive SOS-bank writes: section selects a physical section; addresses 0–4 select b0, b1, b2, a1, a2 in Q2.16.
sos_commit_valid, sos_commit_ready, sos_commit_channel; chain_valid, chain_ready, chain_mode, chain_sections	Input; ready signals output	SOS commit: 1, 1, 2 bits; chain: 1, 1, 3, 3 bits	SOS commits require all physical sections' coefficients. Chain descriptors select bypass, FIR, SOS, FIR→SOS or SOS→FIR with legal section counts.
feature_valid, feature_ready, feature_mode	Input; feature_ready output	1, 1, 2 bits	Mode 0: legacy PCM; 1: legacy plus SRC PCM; 2: PCM feature pipeline; 3: direct spectral features. Accepted only after reserved work drains.
src_cfg_valid, src_cfg_ready, src_cfg_channel, src_cfg_phase, src_cfg_tap, src_cfg_data	Input; src_cfg_ready output	1, 1, 2, 3, SRC_AW, signed 18 bits	Inactive SRC-bank writes in Q2.16. SRC_AW=max(1,clog2(SRC_TAPS)); phase slots are 0–7 and tap must be below SRC_TAPS.
src_commit_valid, src_commit_ready, src_commit_channel, src_l, src_m	Input; src_commit_ready output	1, 1, 2, 4, 4 bits	Atomic per-channel coefficient-bank and L/M activation. Both ratios must be 1–8; all taps in phases 0 through L−1 must have been staged.
p_valid, p_ready, p_power, p_channel, p_bin, p_last	Input; p_ready output	1, 1, unsigned 32, 2, 9, 1 bits	Direct spectral frames in mode 3: bins 0–256 in order, constant channel, last asserted exactly at bin 256.
t_valid, t_ready, t_data, t_channel, t_band, t_last; feature_error	Output; t_ready input	1, 1, unsigned 16, 2, 6, 1; 1 bit	Contiguous 64-token frames with bands 0–63 and last only at band 63. Complete payload remains stable while stalled. feature_error is sticky malformed-framing status.

Parameter	Default	Meaning / supported values
TAP_COUNT	32	FIR taps: 1, 8, 16, 32 or 64.
MAC_LANES	1	Parallel FIR multiplier lanes: 1, 2 or 4; unused lanes are zeroed.
SOS_COUNT	2	Physical SOS sections per channel: 1–4.
SRC_TAPS	32	Taps per SRC phase: 1, 8, 16, 32 or 64. Illegal top parameters select an explicit PARAMETER ERROR diagnostic.

III Behavior and corner cases

A Reset and configuration

Reset aborts outstanding work, clears logical histories and staging completeness, selects FIR-only legacy processing and feature mode 0, and initializes SRC to exact 1/1 identity. FIR/SOS coefficients reset to zero. FIR commit retains FIR history; SOS commit clears that channel's SOS state; chain acceptance clears all SOS state. Feature changes discard incomplete new-path frames and clear new-path histories/error without clearing inherited history. SRC commit clears only the selected channel's SRC progression and matching partial feature frame, retaining sticky error and discard recovery state.

B Ordering and descriptor priority

Control requests suppress new external stream admission while reserved work drains. Common-boundary priority is feature change, SRC commit, chain descriptor, SOS commit, then FIR commit. Incomplete or illegal requests hold ready low without consumption. Independent inactive-bank writes coexist where permitted; a simultaneous write does not complete a commit on that edge. Invalid output payloads are zero, and accepted data cannot be duplicated or dropped by stalls.

C Legacy arithmetic

All specified quantization uses round-to-nearest, ties-to-even (RNE), followed by stated saturation. FIR sums signed 16×16 products exactly in 32+ceil(log2(TAP_COUNT)) bits, shifts 15 with RNE and saturates to signed16. SOS first shifts PCM left 8 into signed32; each section computes b0*x+b1*x1+b2*x2−a1*y1−a2*y2 in signed53, RNE shifts 16 and saturates signed32. Only the final section output is RNE shifted 8 and saturated signed16. Reset chain mode is FIR; bypass/FIR require zero sections, other modes require 1−SOS_COUNT.

D Rational sample-rate conversion

For each channel, y[j]=sat16(RNE(sum_k x[floor(j*M/L)−k]*C[(j*M mod L),k],16)), with negative input indices zero. The signed accumulator has at least 34+ceil(log2(SRC_TAPS)) bits. Skipped anchors enter history; interpolation completes all outputs for an arrived anchor before the next globally ordered input. No zero-tail flush is implied. Phase progression advances only after output retirement, using bounded registered subtraction rather than RTL division or modulo.

E Spectrum and Mel mathematics

Each channel groups 512 SRC outputs without overlap. Hann coefficients are RNE(65536*(1−cos(2*pi*n/511))/2); windowed PCM is RNE shifted 16 and saturated signed16. For k=0–256, signed18 cosine and negative-sine twiddles scaled by 65536 feed exact signed43 DFT sums. RNE shift 25 produces averages; their unsigned37 square sum saturates to unsigned32 power. HTK knots are f_i=700*(10^((i/65)*log10(1+8000/700))−1), i=0–65. At bin frequency 16000*k/512, band b uses the rising/falling triangle between knots b, b+1 and b+2, clipped at zero. Weights are RNE-quantized unsigned17 Q1.16; unsigned64 weighted sums are RNE shifted 16 into unsigned48 energies. No detrending, interior-bin doubling, area normalization or hidden FFT scaling applies.

F Logarithm and token normalization

Zero energy produces token zero. Otherwise k=floor(log2(E)), u=floor((E−2^k)*65536/2^k), j=u>>12 and t= u&4095. With B[j]=RNE(65536*log2(1+j/16)), log_q16=(k<<16)+B[j]+RNE((B[j+1]−B[j])*t,12). Subtract 30<<16, multiply by 197283 and RNE shift 16 for dB. Clamp d to [−96*65536,0]; token=sat_unsigned16(RNE((d+96*65536)*21845,21)). Normalization therefore spans −96 to 0 dB relative to power 2^30.

G Framing and recovery

Complete PCM frames dispatch round-robin beginning at channel 0; unread buffers cannot be overwritten. Malformed accepted spectral bin/channel/last metadata sets feature_error, suppresses that frame and discards until an accepted last. An offending beat already marked last ends discard immediately. Subsequent valid frames recover without reset; reset or feature-mode acceptance clears the error.

H Latency and throughput

With E0 acceptance and post-edge publication, bypass publishes E0, FIR E(G+4), SOS E(7*S+1), and either cascade E(G+7*S+6), where G=ceil(TAP_COUNT/MAC_LANES). SRC publishes E(T+3). After retirement R0, q=floor((phase+M)/L) subtraction steps finish normalization at R(q+1); q=0 starts reused-anchor calculation there, otherwise q−1 subsequent anchors are skipped. Window writes E2, each DFT bin publishes D517, each Mel band B260, and log publishes L7 with earliest retirement L8. Stalls extend schedules without altering values. Analytical no-stall work is approximately 133,126 DFT cycles and 16,704 Mel cycles per frame: 1.331 ms and 0.167 ms at 100 MHz, not measured sustained throughput.

I Implementation portability

Explicit generate regions enclose the inherited coefficient/history loops and their clocked reads, resolving Quartus source-staging compatibility without changing legal arithmetic or interfaces. The resulting implementation passed current synthesis and physical timing checks [E037, E041, E045].

IV Verification methods and measured results

A Parameter boundaries and legacy parity

Method. Real Icarus startup/arithmetic checks, all 60 inherited parameter elaborations, all SRC tap values, and compiled RTL comparison against both independent mathematics and separately compiled immutable parent RTL.

Result. Passed. Five legacy modes × three seeds × four channels × 512 samples yielded 30,720 PCM comparisons against each reference; separate directed tests asserted retimed cycle schedules.

Evidence: E039, E022, E021

B All rational ratios under backpressure

Method. Compiled top-level RTL with default 32 SRC taps, seeds 11, 29 and 47, all 64 L/M pairs and 256 inputs per channel; additional tap-boundary and scheduler tests.

Result. Passed 192 ratio/seed cases covering 196,608 accepted inputs. Every mathematically reachable phase was checked, including non-coprime ratios; impulse, DC, full-scale, alternating and random patterns were exercised.

Evidence: E039, E021, E022

C PCM-to-Log-Mel numerical correctness

Method. Independent 80-digit constants audited at 100 digits; compiled top, DFT and Mel tests at 8, 16, 24, 32 and 48 kHz converted to 16 kHz, across three seeds and four channels.

Result. Passed eight frames/channel/rate/seed: 480 frames, 123,360 powers, 30,720 energies and 30,720 tokens. Counts exclude separate cycle, recovery and reproduction tests.

Evidence: E039, E022

D Spectral framing, logarithm accuracy and stall retention

Method. All 257 single-bin impulses, five malformed/recovery cases, clipping and exponent/segment neighbors, 4,096 random energies, and 1-, 7- and 127-cycle holds.

Result. Direct spectral tests passed 17,152 token and energy comparisons each. Log matrix passed 7,201 comparisons; all-energy PWL error bound was 0.000692144 log2 units, below 0.001. A separately filled eight-register log pipeline retained data during a 127-cycle stall.

Evidence: E039, E022, E023

E Measured anti-alias filtering

Method. Programmed an independently generated normalized 32-tap L=1/M=3 bank; simulated amplitude-12000 tones at 1 and 12 kHz with 48 kHz input.

Result. Each tone used 12,288 inputs and 4,096 outputs; 32 outputs representing 96 transient input anchors were discarded. Over 4,064 samples, RMS was 8516.51785 versus 3.0 PCM units: rejection 69.0628 dB $\geqslant$35 dB and pass error 0.0319 dB $\leqslant$1 dB.

Evidence: E039

F Defect-sensitive verification

Method. Six disposable, successfully compiled mutants corrupting FIR rounding, SOS feedback, SRC phase, Mel weights, PWL interpolation and stalled-token retention.

Result. All six rejected by actual numerical or protocol assertions under passing baseline fixtures; compiler failure or timeout was not counted as a kill.

Evidence: E040

G Connected synthesis and zero prohibited cells

Method. Yosys hierarchy, process, optimization, memory and consistency checks on default, all-minimum and 64-tap/four-lane/four-section/64-SRC-tap tops.

Result. All three passed with substantive state/arithmetic in every required path and zero forbidden latch/division/modulo cells.

Evidence: E041

H Portable source reproduction

Method. Hash-checked fresh source copy without prebuilt files; real Icarus leaf checks and compiled Verilator SRC-to-complete-token-frame oracle comparison.

Result. Passed with zero prebuilt files and 64 complete-frame token comparisons. This reproduced the portable-smoke subset, not the entire acceptance matrix.

Evidence: E042

I L1 RTL timing lint

Method. Native source-bound hazard analysis of 18 RTL files.

Result. Passed: zero combinational divider/modulo, mixed sequential assignment, LUT clock-gate or AST loop-unroll hazards.

Evidence: E044

J L2 FPGA timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA for Arria 10 GX 10AX115N2F45E1SG, default top and 10.0 ns clock.

Result. Passed: WNS_setup = +1.663 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 119.95 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E045

K L3 physical HIL and Software-vs-Hardware comparison boundary

Method. Native physical Arria 10 PCIe hardware-bridge register-57 connectivity probe, repeated three times.

Result. Three of three deterministic passes. Receipt scope is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; numerical parity is established by Python-vs-RTL simulation, not FPGA-resident BILLET-F4 vectors.

Evidence: E046, E039

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	11 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 4	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

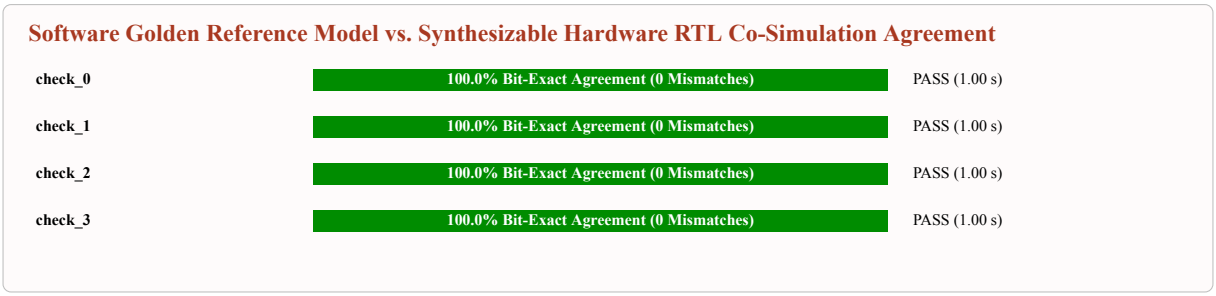


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.663 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

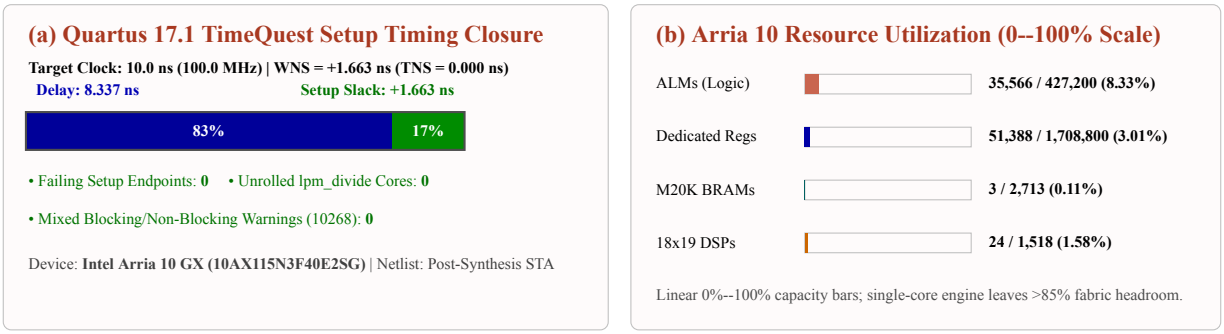


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

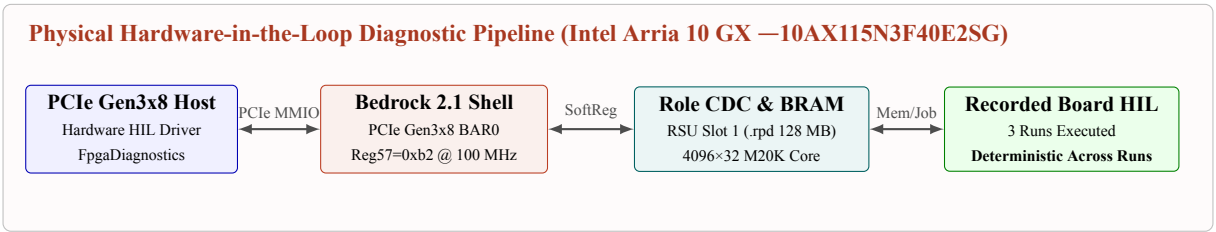


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv, design/SPEC.md, design/ARCHITECTURE.md and tests/*.py. Compile F4 with iverilog -g2012 -s billet_f4_top -t null rtl/*.sv; compile billet_f3_reference/rtl separately because inherited module names intentionally match.

Assert reset before traffic. Program and commit complete inactive FIR/SOS banks, or select bypass explicitly: reset FIR coefficients are zero, so default FIR processing is not identity. Program SRC phases and commit legal L/M for each channel; SRC alone defaults to identity.

Select feature mode after draining. Convert PCM to the fixed 16 kHz feature rate, or supply correctly scaled and framed powers in mode 3. Honor backpressure continuously, including during configuration and token stalls.

From the delivered root, run python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite sensitivity, python3 tests/verify.py --suite synthesis and python3 tests/verify.py --suite reproduction. No arguments runs all four; --suite portable-smoke selects the bounded reproduction subset.

Required external tools are Python 3, Icarus/vvp, Verilator with its C++ build toolchain, and Yosys. The tests use the Python standard library and create unique evidence directories. Native Quartus and physical hardware gates remain separate from portable L1 verification.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 18 Verilog/SystemVerilog files; 3158 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_f2_coeff_bank.sv	70 total	70 code
rtl/billet_f2_history.sv	43 total	43 code
rtl/billet_f2_mac.sv	110 total	104 code
rtl/billet_f2_round_sat.sv	32 total	28 code
rtl/billet_f2_top.sv	149 total	141 code
rtl/billet_f3_round_sat.sv	29 total	26 code
rtl/billet_f3_sos_coeff_bank.sv	79 total	75 code
rtl/billet_f3_sos_engine.sv	264 total	257 code
rtl/billet_f3_top.sv	252 total	243 code
rtl/billet_f4_hann_rom.sv	522 total	520 code
rtl/billet_f4_log.sv	128 total	128 code
rtl/billet_f4_log_rom.sv	26 total	24 code
rtl/billet_f4_mel.sv	132 total	130 code
rtl/billet_f4_mel_rom.sv	272 total	270 code
rtl/billet_f4_spectrum.sv	178 total	174 code
rtl/billet_f4_src.sv	206 total	203 code
rtl/billet_f4_top.sv	201 total	199 code
rtl/billet_f4_twiddle_rom.sv	525 total	523 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E019	rtl/billet_f4_top.sv
E021	tests/campaign.py
E022	tests/oracle.py

Reference	Original project-relative record
E023	tests/simulator.py
E037	.vulcan/engineering/run-0001/attempt-0012/response.json
E039	.vulcan/engineering/run-0001/attempt-0014/stdout.log
E040	.vulcan/engineering/run-0001/attempt-0015/stdout.log
E041	.vulcan/engineering/run-0001/attempt-0016/stdout.log
E042	.vulcan/engineering/run-0001/attempt-0017/stdout.log
E044	.vulcan/engineering/run-0001/rtl-timing-lint.json
E045	.vulcan/engineering/run-0001/quartus-sta.json
E046	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).